

Artificial intelligence-based heritage tree disease diagnosis using transfer learning: A case study of *Zelkova serrata*

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Abstract

Heritage trees are highly valued and protected by national laws because of their cultural and historical significance. However, due to the financial and time constraints associated with the continuous monitoring of the heritage trees, unnecessary losses are reported annually. As a solution, this study suggests an artificial intelligence(AI)-based heritage tree disease diagnosis system on *Zelkova serrata*. We have compared several state-of-the-art deep learning models with transfer learning on the *Zelkova Serrata* Dataset which consists of 680 images. All models achieved outstanding classification results even only using pre-trained weights of ImageNet, with F1 scores ranging from 92.00% to 96.26%. Particularly when additionally leveraging the plant disease datasets, model performances improved to a range of 93.78% to 99.45%. Through this research, we proposed the concept of AI-based heritage tree disease diagnosis using transfer learning. This system is expected to reduce the aforementioned financial and time constraints.

Keyword

Artificial intelligence; Heritage tree; Disease diagnosis; Transfer learning; *Zelkova Serrata*

1. Introduction

The heritage tree is considered as a living cultural asset, representing cultural and traditional aspects of the ecosystem, holding special significance in human society [1]. They coexist with us, often containing legends and regional traditions [2]. This dignified treasure is managed and protected at the national level, with stringent selection criteria

based on factors like size, historical importance, and uniqueness [3-5]. For example, *Zelkova serrata*, also known as Japanese zelkova, is a representative heritage tree in East Asia [6]. Notably, in South Korea, over 50% of government designated protected trees are *Zelkova serrata*, underscoring its cultural significance [7].

However, as shown in Figure 1, heritage trees are susceptible to various forms of damage, such as sudden death and deterioration. Although most heritage trees become resilient over their long lifespans in response to internal and external disruptions [8], this resilience rather makes it hard to identify tree disease-related symptoms [1]. Continuous observation is required for immediate treatment when tree disease-related symptoms appear. Yet, early identifying these symptoms are difficult because there are only a limited number of tree experts and it's uncertain to predict when the tree disease-related symptoms might manifest. Therefore, a technology which can efficiently identify these symptoms is crucial.

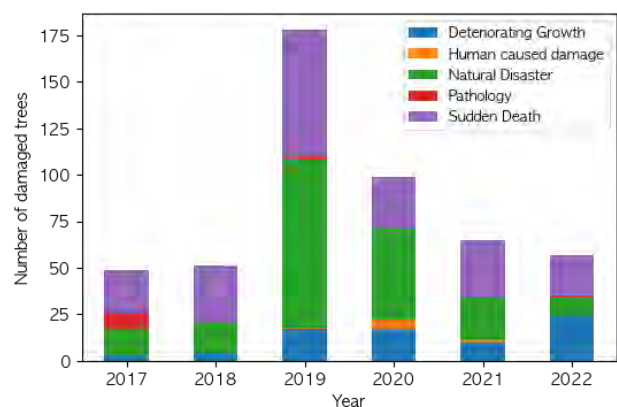


Figure 1. The number of damaged heritage trees by causes in South Korea. Note that the sudden increase in damage in 2019 can be attributed to the influence of Typhoon LINGLING [9].

Recent studies in farming, including orchards, forest areas [10, 11] explored artificial intelligence (AI) technology as a solution to identify ambiguous disease symptoms. Specifically, studies are research to construct relevant datasets and build appropriate AI models [12]. For example, several studies tried to collect data by scraping images from the Internet [13, 14]. In addition, other researchers used transfer learning, which possesses the capability to exhibit respectable performance even with limited data, to build a workable AI model in farming [15]. However, there has been relatively little interest in the problems of heritage tree disease diagnosis, and there is no available dataset for utilization. In this context, AI-based heritage tree disease diagnosis is still an open question.

Taken together, this study aims to propose a setting to maximize AI-based heritage tree disease diagnosis performance and verify its applicability through a case study of *Zelkova serrata*. The paper is structured in the following. In Section 2, we describe the way we collect the two classes (healthy vs. diseased) of the *Zelkova Serrata* Dataset (ZSD). We also describe Plant Disease Datasets (PDD), an already well-refined large datasets similar to heritage trees. Both datasets were augmented by data transformation. In Section 3, we explain our deep learning models and their training settings. In Section 4, we compared each deep learning models' performance on heritage tree disease diagnosis. Finally, in Section 5, we present an overall discussion of our findings, and provide interpretations of the experiment results.

2. Heritage tree disease diagnosis dataset

2.1 Zelkova Serrata Dataset (ZSD)

As mentioned above in Section 1, the absence of a dataset for tree disease diagnosis posed a significant challenge. To address this gap, we constructed our own dataset comprising *Zelkova serrata* leaf images sourced from Google Images,

Shutterstock, and Flickr. It's worth noting that the available images presented certain challenges, including a scarcity of clear representations of *Zelkova serrata* characteristics, along with inherent issues such as randomness, low image quality, and complex backgrounds that could potentially impede the learning process. Consequently, we manually selected images exhibiting relatively clear features of the leaf stem or disease symptoms. Given that many images contained multiple leaves within a single image, we carefully cropped them to isolate individual leaves, as depicted in Figure 2.

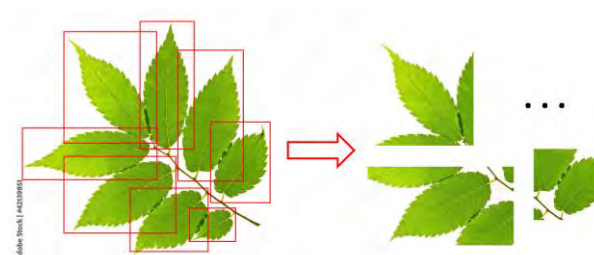


Figure 2. An example image of *Zelkova Serrata* Dataset and its cropped leaves.

This effort yielded a total of 680 images, encompassing both *healthy* ($n = 457$) and *diseased* classes ($n = 223$). Subsequently, we partitioned this *Zelkova Serrata Dataset (ZSD)* into training and test sets, maintaining an 8:2 ratio. In cases where the diagnosis was uncertain due to low resolution, we applied a super-resolution technique, specifically the enhanced deep super-resolution technique [16], exclusively to images without significant distortion at the edges. To ensure the accuracy of labels for these images, validation was performed by a seasoned tree expert with 29 years of experience in tree disease diagnosis.

2.2 Plant Disease Datasets (PDD)

The *Plant Disease Datasets (PDD)* are repositories of open images primarily focused on crops susceptible to diseases, including potatoes, corn, apples, cherries, and more. These datasets serve as valuable resources for leveraging computer vision techniques to distinguish between healthy and

abnormal (i.e., diseased) crops. In previous study [17, 18], an investigation highlighted the significance of domain similarity when applying transfer learning within a specific domain. Essentially, this suggests that greater domain similarity leads to improved AI model performance. Consequently, we anticipate enhanced performance by transferring knowledge from the PDD.

From the array of PDD, as outlined in Table 1, we specifically utilized the PlantVillage [19], PlantDoc [13], and FGVC8 [20] datasets. The rationale behind selecting only these three datasets from the PDD stems from their adherence to the FAIR guiding principles [21]—an acronym signifying ‘Findable,’ ‘Accessible,’ ‘Interoperable,’ and ‘Reusable.’ Within these three datasets (PlantVillage, PlantDoc, and FGVC8), we intentionally selected classes closely resembling the symptoms associated with the *Zelkova serrata* diseases. Classes such as ‘shrub’ or ‘tuber’ were intentionally excluded, and for the purpose of enhancing generalization capabilities, these classes were re-grouped into two classes: healthy vs. diseased.

2.3 Data Transformation

We applied a transformation to both ZSD and PDD, following the criteria suggested by [22]. This transformation encompassed several steps. Initially, for the random resize crop, the pixel size was determined by referencing the pre-trained model’s

image size. Subsequently, random horizontal flips were applied to the training data, ensuring resizing to the same specified pixels as the training data, followed by center cropping. To maintain consistency, each dataset was then normalized using statistics derived from ImageNet [23]. This normalization resulted in mean values of [0.485, 0.456, 0.406] and standard deviation values of [0.229, 0.224, 0.225].

3. Methods

3.1 Models

In the realm of plant disease diagnosis, an essential consideration is the practical applicability of the model at the application level [12]. A recent study [24] compared the performance of cutting-edge deep learning models for plant disease classification. This comparison included Vision Transformer (ViT) [25] and Convolutional Neural Network (CNN)-based models like MobileNet [26], EfficientNet [27], and hybrid models known for their relatively lightweight architecture. The findings revealed that ViT-based models demonstrated superior performance with fewer parameters, albeit slightly slower, while CNNs exhibited the opposite trend. This aligns with observations in various domains [25]. It’s noteworthy that previous studies did not explore ViTs optimized for efficiency, such as MobileViT [28] or efficientFormer [29], suggesting potential for enhanced speed and overall efficiency.

Table 1. Plant Disease Datasets (PDD) that adhere to the FAIR guiding principles.

	PlantVillage	PlantDoc	FGVC8
Included class	Apple, Cherry, Orange, Peach	Apple, Cherry, Peach	Apple
Excluded class	Potatoes, Corn, Squash, Tomato, etc.	Potatoes, Corn, Squash, Tomato, etc.	-
Number of selected images (Healthy / Diseased)	13,187 (2,859 / 10,328)	438 (257 / 181)	18,635 (4,627 / 14,008)
Characteristics of the dataset	Clear background	Complex background	High resolution

Considering both the evidence of ViT outperforming CNNs in addressing plant diseases, which closely relates to our paper's topic on "Heritage Tree Disease" [17], and the potential to harness the improved ViT in terms of efficiency, the utilization of ViT in our research appears to be the best choice.

Taking into account practical utility at the application level, while weighing the benefits in terms of efficiency and memory usage, we acquired four different models—both CNN and ViT-based—from HuggingFace, a repository enabling the download of pre-trained model weights, including ViT. Here's a brief introduction to the selected models:

EfficientNet-B0 [27]: This is a CNN-based model known for its efficiency and speed. It has been frequently employed and validated in previous plant disease research [12].

MobileNet-V2-1.0 [26]: Another CNN-based model, MobileNet-V2-1.0, is optimized for mobile environments and widely used in the field of plant disease detection due to its applicability. It has demonstrated good performance in many cases within the PDD [12]. Thus, this model can be used as an indicator to verify whether the composition of ZSD is appropriate.

EfficientFormer-L1 [29]: This is one of the enhanced versions of ViT, designed for lower latency, allowing for faster inference times. This model has an advantage in scenarios where speed is critical and offers an opportunity for an indirect comparison with EfficientNet.

MobileViT-V2 [28]: Engineered to mitigate the drawbacks of relatively heavier ViT models [25], MobileViT-V2 boasts several advantages, including a lower parameter count compared to MobileNet. This translates to improved memory allocation and makes it suitable for an indirect comparison with MobileNet.

All the models have been configured to produce a 2-class output feature at the final layer of the

model architecture, classifying between "diseased" and "healthy." Training was conducted following the settings described below.

3.2 Training settings

Optimizer: To enhance the model's generalization despite its overparameterization, we employed both AdamW [30] and A-SAM [31].

Scheduler: The learning rate was structured as a step function, wherein it's divided by 0.2 raised to the power of 3/10, 6/10, and 8/10, respectively.

Criteria: To handle uncertainty, we implemented label smoothing [32] with cross-entropy [33], allocating 10% of the probability mass to non-gold classes and 90% to the gold class.

Additional considerations: Monitoring of the training process involved tracking training logs, while the data loader was configured with a batch size of 128 on virtual machines with NVIDIA A100 GPUs.

In cases where we solely employed ImageNet weights for pretraining on the ZSD (Section 2.1), we set the number of epochs to 100 for fine-tuning. When incorporating the PDD (Section 2.2) for pretraining, the epoch count was adjusted to 30. Subsequently, for fine-tuning on the ZSD, we utilized 80 epochs, guided by insights from [27, 29]. Furthermore, to ensure robustness, we applied a 5-fold cross-validation during the training phase.

4. Results

We present varying performance results based on different configurations, including epochs and pretrained weights. As shown in Table 2, results showed the remarkable differences in disease classification performance on the test set when comparing models trained with and without the use of the PDD. Notably, all models trained on ZSD and subjected to transfer learning with PDD displayed improved performance compared to those trained without transfer learning.

Table 2. Comparison results on ImageNet-1K, PDD, and ZSD. The experiments were conducted with two CNN-based models and two ViT-based models.

Model	Type	Param (M)	FLOPs	image size	Pretrained weights	epoch	Test Accuracy(%)	Test F1-score(%)
EfficientNet-B0	Conv	5.3	422 M	224 ²	ImageNet	100(Zelkova)	91.66	92.00
					ImageNet +PDD	30(PDD) / 80(Zelkova)	94.27	93.78
MobileNet-V2-1.0	Conv	3.4	314 M	224 ²	ImageNet	100(Zelkova)	94.26	93.92
					ImageNet +PDD	30(PDD) / 80(Zelkova)	93.17	96.17
EfficientFormer-L1	Meta Block (ViT)	12.3	-	224 ²	ImageNet	100(Zelkova)	96.35	96.26
					ImageNet +PDD	30(PDD) / 80(Zelkova)	99.47	99.45
MobileViT-V2-1.0	Hybrid (ViT + Conv)	4.9	1.8 G	224 ²	ImageNet	100(Zelkova)	95.83	95.50
					ImageNet +PDD	30(PDD) / 80(Zelkova)	98.95	98.91

Specifically, MobileViT-V2-1.0 exhibited a substantial increase in F1 score, soaring from 95.50% to an impressive 98.91% on the test set.

Furthermore, it's worth highlighting that ViT-based models, namely EfficientFormer-L1 and MobileViT-V2-1.0, consistently outperformed CNN-based models like EfficientNet-B0 and MobileNet-V2-1.0 across all settings. An intriguing observation is that MobileViT achieved superior performance despite having fewer parameters than EfficientNet-B0. This trend aligns with previous research findings indicating that ViT-based models can outperform CNN-based models in plant disease detection on PDD [24], even when dealing with limited computational resources [28, 29].

5. Discussion and Conclusion

In this research, we conducted a case study focused on AI-based heritage tree disease diagnosis, addressing the challenge of the unavailability of

suitable datasets for heritage trees. To enhance the performance of heritage tree disease diagnosis, we explored the potential impact of leveraging materials and models from plant disease research, which share similarities with heritage tree diseases [15, 17, 18]. This investigation underscores the potential of applying transfer learning to AI-based heritage tree disease diagnosis.

In terms of materials and models, we pursued transfer learning, taking into account the resemblances between heritage tree diseases and plant diseases. We trained two ViT-based models and two CNN-based models with an emphasis on efficiency and lightweight design on both PDD and ZSD datasets [26-29]. Our findings indicate that leveraging the PDD, which shares similarities with ZSD, led to a significant improvement in disease diagnosis performance. This suggests that knowledge transfer from plant disease-related symptoms to heritage tree disease-related symptoms is effective [15, 17]. Consequently, the

potential of using PDD to enhance AI-based heritage tree disease diagnosis performance is a promising avenue for future research in this field. This characteristic may offer valuable insights for future studies on AI-based heritage tree disease diagnosis by considering the properties of the target domain.

Regarding our experimental results, we posit that the superior performance of ViT-based models compared to CNN-based models is not solely attributed to the number of parameters. The fact that MobileViT-V2-1.0, with relatively fewer parameters, outperforms EfficientNet-B0 with more parameters, suggests that the ViT structure itself is better suited for heritage tree disease diagnosis, emphasizing the structural aspect over parameter count [28, 29]. Considering the ongoing development of high-performing ViT-based model architectures [34] and our findings regarding their outstanding performance in heritage tree disease diagnosis, employing optimized ViT-based models could be a promising strategy for future related research.

However, our study has certain limitations. The dataset lacks sufficient clear background images captured from the same angle since we sourced data from the internet. This may pose challenges for deep learning models to capture leaf characteristics effectively, indicating the need to incorporate both clear and complex background images to enhance the training process [13, 15]. Additionally, environmental factors surrounding heritage trees were not considered in our study. Future research should aim to incorporate the overall context of the tree to develop a more comprehensive AI-based heritage tree disease diagnosis system. Consequently, our future studies will explore multi-modal AI-based heritage tree disease diagnosis, integrating more images and environmental data with a refined dataset [35].

Nonetheless, this study unveils the potential for AI to address the ambiguities associated with heritage tree disease diagnosis as pointed out in the introduction. To substantiate this proposition, we

demonstrate that transfer learning can enhance AI-based heritage tree disease diagnosis performance by leveraging the PDD. Furthermore, we propose ViT-based models as the more suitable choice through a comparative analysis of several frequently used state-of-the-art deep learning models. In summary, our findings suggest that AI-based heritage tree disease diagnosis can offer solutions to long-standing questions in this field, and we anticipate continued exploration in this area of research.

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